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Characterization of ketolactia in dairy cows during early lactation

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ABSTRACT

Fourier transform infrared spectroscopy (**FTIR**) allows for the determination of milk acetone (mACE) and β-hydroxybutyrate (mBHB) concentrations, providing a potential herd monitoring tool for hyperketolactia, defined as elevated milk ketone bodies. The study aim was to characterize mACE and mBHB concentration dynamics during early lactation in Polish Holstein-Friesian cows. Milk samples (n = 3,867,390) were collected within 6 to 60 days in milk (DIM) over a 4-yr period (April 1, 2013 to March 31, 2017) from approximately 21,300 dairy herds (average 38.7 cows/herd). Fixed effects of parity, DIM, and their interaction on mACE and mBHB concentrations were determined using a mixed model with a herd-year-season fixed effect and random cow effect. Published hyperketolactic mACE (≥0.15 mmol/L) and mBHB (≥0.10 mmol/L) threshold concentrations were used to classify study milk samples into ketolactia groups of normal (mACE <0.15 mmol/L and mBHB <0.10 mmol/L) and hyperketolactic (HYKL; either mACE ≥0.15 mmol/L or mBHB ≥0.10 mmol/L). Additionally, HYKL samples were categorized into subpopulations as having elevated mBHB and mACE ($HYKL_{ACEBHB}$, mACE ≥0.15 mmol/L and mBHB ≥0.10 mmol/L), only elevated mBHB ($HYKL_{BHB}$; mACE <0.15 mmol/L and mBHB ≥0.10 mmol/L), or only elevated mACE ($HYKL_{ACE}$; mACE ≥0.15 mmol/L and mBHB <0.10 mmol/L). Effects of parity, DIM, ketolactia group or subpopulation, and their interactions on mACE and mBHB concentrations were also determined using the mixed model that included ketolactia group or subpopulation as an independent variable. Across all data, mACE and mBHB concentrations were influenced by effects of parity, DIM, and their interaction as well as parity, DIM, ketolactia group or subpopulation, and their interactions. For all samples, mACE and mBHB concentrations decreased with increasing DIM, with mACE concentration declining more rapidly compared with mBHB. In the data set, 68% and 32% of all samples were defined as normal or HYKL, respectively. Among HYKL samples, mACE was elevated soon after calving and declined over time. In contrast, mBHB started lower after calving and increased reaching peak concentrations around 30 DIM, and then decreased. Within HYKL samples, 50.8, 41.3, and 7.9% were categorized as $HYKL_{ACEBHB}$, $HYKL_{BHB}$, and $HYKL_{ACE}$ respectively. Between 6 and 21 DIM, 11.3% of HYKL were classified as $HYKL_{ACE}$. Primiparous cows had greater (14.8%) $HYKL_{ACE}$ samples in this time period. In conclusion, this study has characterized mACE and mBHB concentrations during early lactation and determined effects of parity, DIM, and their interaction. Using published criteria interpreting mACE and mBHB concentrations, it was intriguing to identify a unique population of samples having elevated mACE without mBHB in early lactation, especially in primiparous cows. Further research is needed to determine if this sample population represents an unhealthy metabolic status that adversely affects cow health and performance.

Key words: hyperketolactia, milk, acetone, β-hydroxybutyrate, Fourier transform infrared spectroscopy

INTRODUCTION

Hyperketonemia is a metabolic disorder characterized by an abnormally high concentration of circulating ketone bodies [i.e., acetoacetate (**AcAc**), BHB, and acetone (**ACE**)] in blood, urine, and milk (Andersson and Emanuelson, 1985). Hyperketonemia is a metabolic

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maladaptive state of excessive negative energy balance and is a primary determinant of subclinical and clinical manifestations of dairy cattle ketosis (Duffield et al., 2009; McArt et al., 2012; Benedet et al., 2019). Hyperketonemia prevalence ranges from 10 to 60% (Suthar et al., 2013; Santschi et al., 2016; Chandler et al., 2018; Benedet et al., 2019), and imparts significant costs (McArt et al., 2015), mostly due to decreased production, impaired health, and reproduction (Duffield et al., 2009; Ospina et al., 2010; McArt et al., 2012). Therefore, effective treatment based on early diagnosis is essential, making reliable diagnostic procedures crucial to managing hyperketonemia.

Urine or milk AcAc concentration can be measured using the nitroprusside reaction; however, neither AcAc nor ACE are sufficiently stable to be measured in blood (Työppönen and Kauppinen, 1980; Fritzsche et al., 2001). Thus, blood BHB concentration is routinely used in assessing a hyperketonemic state, also in cowside tests using hand-held blood ketone meters (Iwersen et al., 2009). Unlike the case in blood, ACE is stable in milk and can be determined by Fourier-transform infrared (**FTIR**) spectroscopy (Hansen, 1999; Heuer et al., 2001; de Roos et al., 2007). This analytical technique has been developed for use in milk recording systems to measure milk components, including milk BHB (**mBHB**) and ACE (**mACE**) concentrations (de Roos et al., 2007; van Knegsel et al., 2010; van der Drift et al., 2012). Determination of mACE and mBHB makes milk analysis a potential herd monitoring tool for hyperketolactia, defined as elevated ketone bodies in milk, and potentially hyperketonemia (Emery et al., 1968; Pralle and White, 2020).

Among 3 main ketone bodies formed in the cow's liver, AcAc is the parent ketone body, and BHB and ACE are its derivatives (Bergman, 1971). The AcAc is reduced to BHB in a reversible reaction catalyzed by hepatocyte cytosolic BHB dehydrogenase. In contrast, AcAc is readily converted to ACE and CO_2 by an irreversible, spontaneous, and nonenzymatic decarboxylation (Bergman, 1971). Metabolic factors directing AcAc conversion into ACE or BHB are not known. An elevated AcAc-to-BHB ratio is indicative of a ketotic state (Andersson, 1984). There might exist cows with elevated blood ACE, given spontaneous ACE formation from AcAc, but without elevations in BHB. Whether this situation would result in adverse cow outcomes would need to be determined. Because FTIR technology provides mACE determination, and mACE is well correlated with blood ACE (Andersson, 1984; Enjalbert et al., 2001), we propose that mACE concentration might be used as a diagnostic tool.

The study objectives were to describe ACE and BHB concentrations dynamics in milk collected from early-lactation Polish Holstein-Friesian cows across parities and characterize cow ketolactia status via milk ketone concentrations. The authors believe this is the first study characterizing ketolactia over the first 60 d of lactation in such a large Holstein population.

MATERIALS AND METHODS

This study was a retrospective survey of milk composition data obtained from the Polish Federation of Cattle Breeders and Dairy Farmers, which provides milk recording in Poland. Data on mBHB and mACE concentrations were collected within the ketosis monitoring system, which was introduced in Poland in 2013, as a part of milk recording (Kowalski et al., 2015). The data set consisted of 3,867,390 milk samples collected over a 4-yr period (April 1, 2013 to March 31, 2017), from approximately 21,300 dairy herds (from 1–1,356 cows per herd, average 38.7 cows/herd; Table 1). Milk samples from only Holstein-Friesian cows were considered in this study, accounting for approximately 95% of Polish milk recorded cows. Because the study covered a 4-yr period, and 2 test-days in the 2 first months of lactation are considered (6–60 DIM) in the system of monitoring of ketosis in Poland, an individual cow could potentially have between 1 and 8 milk samples in the dataset.

All milk samples were collected on test days by trained sampling technicians as part of the herd's routine monthly DHI testing approved by an International Committee on Animal Recordings (**ICAR**). Milk samples were collected using a proportional sampler approved by ICAR, preserved immediately with 2-bromo-2-nitropropane-1,3-diol (bronopol) tablets, and analyzed for chemical composition in 4 Polish Federation of Cattle Breeders and Dairy Farmers laboratories using 9 MilkoScan FT 6000 units (Foss Analytical A/S). All milk samples were gently preheated to 40°C and mixed before analysis in accordance with the instrument manufacturer's instructions. The FTIR instruments were standardized monthly following ICAR guidelines for routine milk components, including ACE and BHB concentrations. The instruments were calibrated for mACE and mBHB determination according to de Roos et al. (2007). Samples with abnormal values of any milk parameter were excluded from the data set. Abnormal values were defined as inability to identify herd, no data on milk composition, protein content <2 or >7%, fat content <1 or >12%, lactose content <3 or >6%, urea content <1 or >999 mg/L, and SCC $<1 \times 10^3$ or $>33 \times 10^6$/mL. Originally, our data set consisted of 4,104,812 samples with 237,422 milk samples excluded. The most common reasons for sample exclusion were abnormal protein or urea content (62.3 and

Table 1. Characteristics of Polish Holstein-Friesian cows and milk samples collected in the 4-yr period from April 1, 2013, to March 31, 2017

Item[1]	Number	Mean	Median	Minimum	Maximum	SD	CV
Number of milk samples							
All	3,867,390						
From first-lactation cows	1,281,123						
From second-lactation cows	983,830						
From third-or-greater-lactation cows	1,602,437						
From cows within 6–21 DIM	1,199,988						
From cows within 22–60 DIM	2,667,402						
Cows characteristics on test-days							
Number	1,283,104						
Parity		2.59	2	1	20	1.72	66.5
DIM		32.33	33	6	60	16.04	49.6
Milk (kg/d)		30.99	30.20	0.20	99.80	9.17	29.6
Number of milk samples per cow		3.01	3	1	11	1.66	55.0
Milk composition							
Protein (%)		3.11	3.06	2.00	7.00	0.36	11.5
Fat (%)		4.12	3.99	1.00	12.00	0.96	23.4
Lactose (%)		4.83	4.85	3.00	5.91	0.22	4.6
Urea (mg/L)		195	189	1	999	85	43.5
SCC ($\times 10^3$ cells/mL)		487	101	1	32,930	1,282	263.5
Fat/protein		1.33	1.29	0.22	5.12	0.31	23.2
mACE (mmol/L)		0.098	0.050	0.000	4.910	0.164	167.3
mBHB (mmol/L)		0.081	0.060	0.000	3.840	0.099	122.0

[1]mACE = milk acetone; mBHB = milk β-hydroxybutyrate.

8.4% of discarded samples, respectively), and difficulty in assigning the milk sample to a specific herd (26.4%). Some samples were discarded for more than one reason.

Ketolactia Definitions

Published mACE (≥0.15 mmol/L) and mBHB (≥0.10 mmol/L) threshold concentrations considered to reflect a hyperketolactic state were used to classify milk samples into ketolactia groups de Roos et al. (2007). Using these threshold values, all samples at any stage of lactation were characterized as nonelevated ketolactia (**NKL**; mACE <0.15 mmol/L and mBHB <0.10 mmol/L) or hyperketolactic (**HYKL**; either mACE ≥0.15 mmol/L or mBHB ≥0.10 mmol/L). Within HYKL samples, there were subpopulations classified as hyperketolactic from both ACE and BHB (**$\mathbf{HYKL_{ACEBHB}}$**; mACE ≥0.15 mmol/L and mBHB ≥0.10 mmol/L), hyperketolactic from only BHB (**$\mathbf{HYKL_{BHB}}$**; mACE <0.15 mmol/L and mBHB ≥0.10 mmol/L), and hyperketolactic from only ACE (**$\mathbf{HYKL_{ACE}}$**; mACE ≥0.15 mmol/L and mBHB <0.10 mmol/L). Additionally, milk samples were evaluated relative to ketolactia within 2 time periods, 6 to 21 and 22 to 60 DIM. This classification of hyperketolactia within these time periods was inspired by Oetzel's (2007) designation of type 1 and 2 ketosis and reported observations of greater prevalence of subclinical ketosis leading to clinical ketosis with adverse effects on cow health and performance (Duffield et al., 2009; McArt et al., 2012; 2015).

Statistical Analysis

Descriptive statistics for milk components, including mACE and mBHB concentrations, were calculated using the MEANS procedure of SAS (version 9.2; SAS Institute Inc) and R environment (https://www.r-project.org/). The R package ggplot2 (Wickham, 2016) was used to create all figures.

Univariate analyses of mACE and mBHB were performed with the following mixed model in ASReml (Gilmour et al., 2015):

$$y_{ijkl} = HYS_i + P_j + DIM_k + P_j \times DIM_k + C_l + \varepsilon, \quad [1]$$

where y = mACE or mBHB; HYS = the fixed effect of herd-year-season of calving (i = 1–290,485); P = the fixed effect of parity (j = 1, 2, ≥3); DIM = the fixed effect of DIM (k = 6–60); C = the random effect of cow (l = 1–1,283,104), proportional to $N\left(0,\mathbf{I}\sigma_c^2\right)$, where σ_c^2 is the variance of cow effects; and ε = residual error. The PREDICT statement of ASReml was used to generate least squares means (**LSM**) for the effects of parity, DIM, and their interaction.

A second series of 2 analyses was performed with either ketolactia group (NKL vs. HYKL) or subpopulation (NKL, $\mathrm{HYKL_{ACEBHB}}$, $\mathrm{HYKL_{BHB}}$, $\mathrm{HYKL_{ACE}}$) added as an independent effect. The linear model of the observation was as follows:

$$y_{ijkl} = HYS_i + H_m + P_j + DIM_k + H_m \times P_j \times DIM_k + C_l + \varepsilon, \quad [2]$$

where H_m = fixed effect of either ketolactia group or subpopulation, and all other effects are as in model [1]. The effect of ketolactia group or subpopulation and the interaction of ketolactia group or subpopulation with parity × DIM was added with LSM derived with the PREDICT statement for ketolactia group or subpopulation × parity × DIM. Because there were HYS subclasses where all cows were normal, HYS was treated as a random effect, and a fixed effect of herd was added to facilitate estimation of LSM. For the test of whether mACE and mBHB are different for the 2-group analysis (NKL vs. HYKL), the P-value from ASReml F-test ($P < 0.001$ with a Kenward-Roger adjustment to the denominator degrees of freedom) was used. In any multiple comparisons performed, a t-test with Bonferroni adjustment was used for LSM multiple comparisons. In this adjustment, the accepted significance ($P \leq 0.05$) was revised by the number of comparisons ($P < 0.05/n$). Comparison number accounted for 2 or 4 ketolactia groups or subgroups, 3 parities, and 55 DIM for a total of 330 or 660 comparisons. For multiple comparisons within DIM periods (6–21 vs. 22–60) a total of 12 (2 groups, 3 parities, 2 periods) or 24 (4 groups, 3 parities, 2 periods) comparisons were used. Data are presented as LSM ± standard error of the mean, except where noted.

Given the large data set, each DIM was considered a class variable, allowing for characterization of change in mACE and mBHB over time. Specific day-to-day mean mACE and mBHB comparisons were not of interest, except where noted, but change in these concentrations over the study time period (6–60 DIM) or within 2 periods (6–21 or 22–60 DIM) were of primary interest. To this end, linear and quadratic regression models of ANOVA-generated LSM of mACE and mBHB by DIM were calculated using SAS. Linear and quadratic regression coefficients were computed within each ketolactia group or subpopulation × parity subclass. Because regression analyses were performed using LSM, P-values were used to help derive the polynomial fit and were not representative of the fit to the entire data set. Similarly, model R^2 values provided a good snapshot as to how well regression lines could fit mean data and assess whether linear or quadratic models were better fits for the data.

For calculating binomial 100(1 − α)% confidence interval we used the Clopper-Pearson exact confidence interval (Clopper and Pearson, 1934; Stevens, 1950) which can be written as follows:

$$\left[\beta_{\alpha/2,m,n-m+1}, \beta_{1-\alpha/2,m+1,n-m}\right], \quad [3]$$

where n = the total sample size, m = the number of successes, and $\beta_{q,a,b}$ = the quantile of order q of the beta distribution $\beta(a,b)$.

RESULTS

Milk Sample Demographics

The study was conducted on 3,876,390 milk samples (Table 1). Approximately 33.1% (n = 1,281,123), 25.5% (n = 983,830), and 41.4% (n = 1,602,437) of milk samples were from primiparous, second-lactation, and third-or-greater-lactation cows, respectively (Table 1). There were 1,199,988 and 2,667,402 records from all cows within 6 to 21 and 22 to 60 DIM, respectively. Mean (± SD) number of milk samples per cow was 3.0 ± 1.7 (median: 3.0; range 1–11). Overall raw data mean mACE and mBHB concentrations were 0.098 ± 0.164 mmol/L (median 0.050; range: 0.00–4.91) and 0.081 ± 0.099 mmol/L (median 0.060; range: 0.00–3.84), respectively. Considering all milk samples in the data set, parity, DIM, and their interaction influenced mACE and mBHB concentrations (all $P < 0.001$).

Parity Effects

Samples of second-lactation cows had the lowest mean mACE (0.097 ± 0.001 mmol/L) compared with first (0.122 ± 0.001 mmol/L) and third-or-greater- (0.118 ± 0.001 mmol/L) lactation cows (all $P < 0.001$). Milk samples of third-or-greater-lactation cows had the highest mean mBHB (0.105 mmol/L ± 0.001) compared with first- (0.093 ± 0.001 mmol/L) and second- (0.092 ± 0.001 mmol/L) lactation cows (all $P < 0.001$).

Time Effects

Mean mACE and mBHB concentrations across parities showed a curvilinear relationship relative to DIM ($P < 0.001$). Both mACE and mBHB concentrations decreased with increasing DIM, though mACE concentration declined more rapidly compared with mBHB (Figure 1). Models regressing DIM on least squared mACE and mBHB means for all samples to describe changing concentration over time showed linear and quadratic DIM effects, with quadratic regression models showing slightly higher R^2 fits compared with linear models for both (Table 2).

Mean mACE concentration for all parities over 6 to 21 DIM and 22 to 60 DIM were 0.157 ± 0.001 (LSM ± SEM) mmol/L and 0.094 ± 0.001 mmol/L, respectively

Table 2. R^2 fits of linear and quadratic regression models of LSM milk acetone (mACE) and β-hydroxybutyrate (mBHB) concentrations on DIM (n = 54), depending on parity and ketolactia group or subpopulation; all models were significant ($P < 0.01$) unless where noted NS

	mACE		mBHB	
Sample group[1]	Linear	Quadratic	Linear	Quadratic
All data				
Parity 1	0.86	0.97	0.97	0.98
Parity 2	0.96	1.00	0.88	0.94
Parity ≥3	0.98	1.00	0.89	0.96
Mean	0.94	0.99	0.97	0.98
Analysis with ketolactia group effect: NKL vs. HYKL				
NKL				
Parity 1	0.84	0.96	0.94	0.95
Parity 2	0.84	0.94	0.36	0.60
Parity ≥3	0.89	0.97	0.44	0.69
Mean	0.87	0.97	0.68	0.79
HYKL				
Parity 1	0.98	0.99	0.65	0.93
Parity 2	0.98	0.99	0.38	0.88
Parity ≥3	0.99	0.99	0.40	0.91
Mean	0.99	0.99	0.48	0.92
Analysis with ketolactia subpopulation effect: $HYKL_{ACEBHB}$ vs. $HYKL_{BHB}$ vs. $HYKL_{ACE}$				
$HYKL_{ACEBHB}$				
Parity 1	0.97	0.97	0.82	0.82
Parity 2	0.97	0.98	0.29	0.81
Parity ≥3	0.98	0.99	0.44	0.87
Mean	0.99	0.99	0.66	0.89
$HYKL_{BHB}$				
Parity 1	0.98	0.98	0.08	0.90
Parity 2	0.91	0.95	0.00[NS]	0.88
Parity ≥3	0.96	0.98	0.00[NS]	0.91
Mean	0.97	0.98	0.00[NS]	0.91
$HYKL_{ACE}$				
Parity 1	0.76	0.93	0.74	0.85
Parity 2	0.62	0.85	0.57	0.61
Parity ≥3	0.78	0.92	0.51	0.62
Mean	0.76	0.94	0.75	0.84

[1]NKL: mACE <0.15 mmol/L and mBHB <0.10 mmol/L; HYKL: mACE ≥0.15 mmol/L or mBHB ≥0.10 mmol/L; $HYKL_{ACEBHB}$: mACE ≥0.15 mmol/L and mBHB ≥0.10 mmol/L; $HYKL_{BHB}$: mACE <0.15 mmol/L and mBHB ≥0.10 mmol/L.; $HYKL_{ACE}$: mACE ≥0.15 mmol/L and mBHB <0.10 mmol/L.

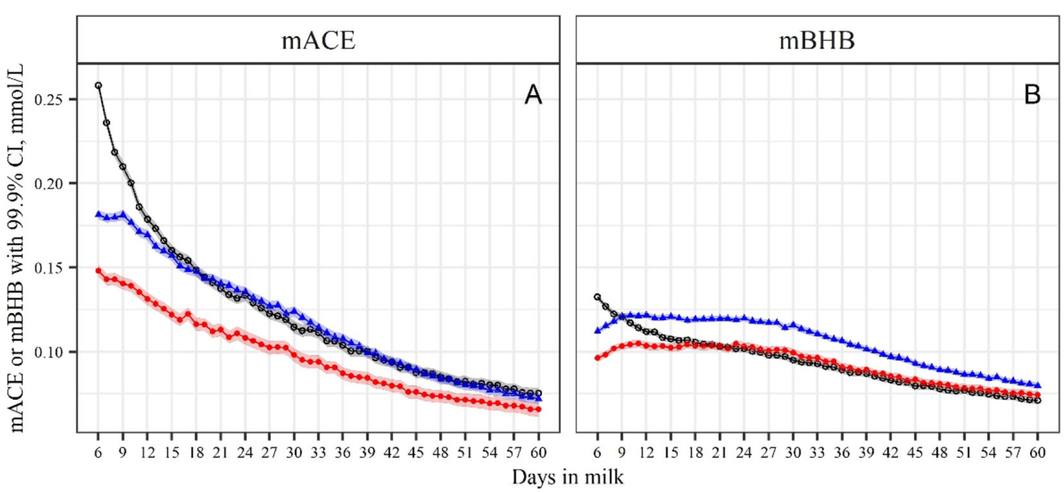

Figure 1. Least squares means of milk acetone (mACE; A) and β-hydroxybutyrate (mBHB; B) concentrations by parity (○ 1, ● 2, ▲ ≥3) and DIM (6–60 DIM; n = 3,863,390 milk samples). Final model included effects of parity ($P < 0.001$), DIM ($P < 0.001$), and a parity × DIM interaction ($P < 0.001$). Shaded area represents 99.9% CI for each DIM within each parity.

($P < 0.001$). Similarly, mean mBHB concentration was higher (0.112 ± 0.06 mmol/L) during 6 to 21 DIM compared with 22 t 60 DIM (0.091 ± 0.001 mmol/L, $P < 0.001$).

Parity × Time Effects

Changes in mACE and mBHB concentrations by parity group over early lactation in Polish Holstein dairy cows are presented in Figure 1. The highest mACE concentration was found in samples originating from primiparous cows in the first few weeks of lactation (Figure 1A). Mean mACE concentration for 6 to 10 DIM was higher in primiparous (0.225 ± 0.001 mmol/L) cows, followed by third-or-greater-lactation (0.180 ± 0.001 mmol/L) cows, with lowest concentration observed in second parity (0.143 ± 0.001 mmol/L) cows (all $P < 0.001$). All parity groups showed a curvilinear decline in mACE concentration over time. Primiparous and third-or-greater-lactation cows had similar mACE concentration from 15 to 60 DIM, whereas second-lactation cows had lower concentrations until 30 DIM ($P < 0.05$), and numerically lower until 60 DIM (Figure 1A).

Beyond 10 DIM, mBHB concentration was highest in third-or-greater-lactation cows compared with other parities (Figure 1B). Averaged LSM mBHB concentration in third-or-greater-lactation cows plateaued at 0.119 ± 0.001 mmol/L from 10 to 30 DIM, then linearly declined through 60 DIM to 0.080 ± 0.001 mmol/L. In contrast to mACE observations, mBHB concentrations in primiparous and second-lactation cows converged around 20 DIM and declined similarly to 60 DIM. From 6 to 10 DIM, mean mBHB concentration was highest in milk of primiparous cows (0.124 ± 0.001 mmol/L), followed by third-or-greater-lactation (0.118 ± 0.001 mmol/L) cows and lowest in second-lactation (0.101 ± 0.001 mmol/L) cows (all $P < 0.05$).

Within 6 to 21 DIM and 22 to 60 DIM periods, mean mACE concentrations were 0.179 ± 0.001 mmol/L and 0.099 ± 0.001 mmol/L, respectively, in primiparous cows; 0.128 ± 0.002 mmol/L and 0.084 ± 0.001 mmol/L, respectively, in second-lactation cows; and 0.162 ± 0.001 mmol/L and 0.100 ± 0.001 mmol/L, respectively, in third-or-greater-lactation cows. Within parity groups, 6 to 21 DIM mACE means were different from 22 to 60 DIM means (all $P < 0.001$). With 6 to 21 DIM, all parity means were different ($P < 0.05$), whereas second-lactation cows had lower mACE compared with primiparous and third-or-greater-lactation cows ($P < 0.05$) within 22 to 60 DIM.

Similarly, mean mBHB concentrations for 6 to 21 DIM and 22 to 60 DIM were 0.113 ± 0.001 mmol/L and 0.085 ± 0.001 mmol/L, respectively, in primiparous cows; 0.103 ± 0.001 mmol/L and 0.088 ± 0.001 mmol/L, respectively, in second-lactation cows; and 0.119 ± 0.001 mmol/L and 0.100 ± 0.001 mmol/L, respectively, in third-or-greater-lactation cows. Within parity groups, 6 to 21 DIM mBHB means were different from 22 to 60 DIM means (all $P < 0.001$). Cows with lactation ≥3 had higher mBHB in periods 6 to 21 DIM and 22 to 60 DIM compared with primiparous and second-lactation cows ($P < 0.05$).

Ketolactia Status Comparisons

In the data set, 67.7 and 32.3% of all samples were defined as NKL and HYKL, respectively, based on described criteria (Table 3). Percent of samples defined as HYKL in 6 to 21 DIM was greater than 22 to 60 DIM samples (43.0 vs. 27.6%, respectively). Overall mean mACE and mBHB concentrations were 0.047 ± 0.001 mmol/L and 0.049 ± 0.0006 mmol/L, respectively, for NKL group cows, and 0.201 ± 0.002 mmol/L and 0.177 ± 0.001 mmol/L, respectively, for HYKL group cows (all $P < 0.001$). Parity, DIM, ketolactia groups (NKL vs. HYKL), and all interactions influenced mACE and mBHB concentrations (all $P < 0.001$).

Parity Effects

Within 6 to 60 DIM, HYKL samples originated from 30.2, 28.2, and 35.4% of first-, second-, and third-or-greater-lactation cows, respectively. The HYKL samples were more prevalent among older cows (lactation ≥3), with the lowest percentage among samples from second-lactation cows (Table 3).

Mean mACE and mBHB in NKL milk samples originating from first-, second-, and third-or-greater- lactation cows were 0.051 ± 0.001 and 0.047 ± 0.001, 0.042 ± 0.001 and 0.049 ± 0.001, and 0.046 ± 0.001 and 0.050 ± 0.001 mmol/L, respectively. In comparison, mean mACE and mBHB in HYKL milk samples originating from first-, second-, and third-or-greater-lactation cows were 0.224 ± 0.002 and 0.167 ± 0.001, 0.176 ± 0.002 and 0.165 ± 0.001, and 0.201 ± 0.001 and 0.177 ± 0.001 mmol/L, respectively. All mean comparisons by parity group between NKL and HYKL status were significant ($P < 0.001$).

Time Effects

For NKL samples across parity groups, mean mACE concentration decreased slightly with increasing DIM (Figure 2A). Mean mACE concentrations were numerically higher over 6 to 14 DIM (0.057 ± 0.001 mmol/L), declining slightly to 0.040 ± 0.001 mmol/L at 60 DIM. This early elevation resulted in a slightly better fit for a quadratic ($R^2 = 0.97$) versus linear ($R^2 = 0.87$) regres-

Table 3. Defined hyperketolactia status in milk samples collected in the 4-yr period from April 1, 2013 to March 31, 2017 from Polish Holstein-Friesian cows

Item	DIM		
	6–60	6–21	22–60
Number of milk samples	3,867,390	1,199,988	2,667,402
NKL, % of all samples[1]	67.7	57.0	72.4
HYKL, % of all samples[2]	32.3	43.0	27.6
From first-lactation cows, % of all from first-lactation cows	30.2	43.9	24.1
From second-lactation cows, % of all from second-lactation cows	28.2	36.2	24.6
From ≥third-lactation cows, % of all from ≥third-lactation cows	35.4	46.4	32.1
$HYKL_{ACEBHB}$[3]			
% of all samples	16.4	26.4	12.0
% of all hyperketolactic due to mACE or mBHB	50.8	61.3	43.4
$HYKL_{BHB}$[4]			
% of all samples	13.3	11.8	14.1
% of all hyperketolactic due to mACE or mBHB	41.3	27.4	51.0
$HYKL_{ACE}$[5]			
% of all samples	2.6	4.8	1.5
% of all hyperketolactic due to mACE or mBHB	7.9	11.3	5.6

[1]NKL: milk acetone (mACE) <0.15 mmol/L and milk β-hydroxybutyrate (mBHB) <0.10 mmol/L.

[2]HYKL: mACE ≥0.15 mmol/L or mBHB ≥0.10 mmol/L.

[3]$HYKL_{ACEBHB}$: mACE ≥0.15 mmol/L and mBHB ≥0.10 mmol/L.

[4]$HYKL_{BHB}$: mACE <0.15 mmol/L and mBHB ≥0.10 mmol/L.

[5]$HYKL_{ACE}$: mACE ≥0.15 mmol/L and mBHB <0.10 mmol/L.

sion model over time (Table 2). Although the effect of DIM on mBHB was significant, it remained relatively stable within 6 to 60 DIM in NKL samples (Figure 2B). Regression modeling of mBHB in NKL samples showed lower fit statistics for linear ($R^2 = 0.68$) and quadratic ($R^2 = 0.79$) models (Table 2).

Among HYKL samples, mean mACE concentrations were initially high (0.312 ± 0.001 mmol/L) and declined to 0.125 ± 0.002 mmol/L by 60 DIM (Figure 2C). Both linear and quadratic models equally fit mACE change over time in HYKL group samples (Table 2). In contrast, mean mBHB concentration started lower after calving and increased, reaching peak concentrations around 30 DIM, and then decreased by 60 DIM (Figure 2D). This curvilinear response over time was best fit with a quadratic model ($R^2 = 0.92$, $P < 0.001$; Table 2). Mean HYKL mACE and mBHB concentrations were greater during the period of 6 to 21 DIM (0.264 ± 0.001 and 0.175 ± 0.001 mmol/L) compared with 22 to 60 DIM (0.174 ± 0.002 and 0.168 ± 0.001 mmol/L), respectively (all $P < 0.001$). Mean mACE and mBHB were greater in the HYKL group compared with NKL group for the periods of 6 to 21 DIM and 22 to 60 DIM (all $P < 0.001$).

Parity × Time Effects

Milk samples of HYKL primiparous cows (0.224 ± 0.002 mmol/L) always had the highest mACE, and HYKL second-lactation cows (0.176 ± 0.002 mmol/L) had the lowest mACE (Figure 2C; $P < 0.05$). Sample mBHB concentration for third-or-greater-lactation cows (0.177 ± 0.001 mmol/L) was always numerically greater than first- (0.167 ± 0.001 mmol/L) or second-lactation (0.165 ± 0.001 mmol/L) cows, excluding the 6 to 9 DIM period (Figure 2D).

Although 22 to 60 DIM HYKL sample percentage did not differ between first- and second-lactation cows, a higher percentage of 22 to 60 DIM HYKL samples originated from third-or-greater-lactation cows (32%) compared with first- (24.1%) and second- (24.6%) lactation cows. Within periods 6 to 21 DIM and 22 to 60 DIM, mean mACE concentrations in NKL samples were 0.059 ± 0.001 and 0.048 ± 0.001 mmol/L for first-lactation cows; 0.049 ± 0.001 and 0.040 ± 0.001 mmol/L for second-lactation cows; and 0.055 ± 0.001 and 0.043 ± 0.001 mmol/L for third-or-greater-lactation cows. In these same 2 periods, NKL mBHB concentrations by parity groups were not different. Mean mACE for HYKL samples by parity group within 6 to 21 DIM and 22 to 60 DIM were 0.303 ± 0.001 and 0.192 ± 0.002 mmol/L for first-lactation cows; 0.230 ± 0.002 and 0.154 ± 0.002 mmol/L for second-lactation cows; and 0.261 ± 0.001 and 0.176 ± 0.001 mmol/L for third-or-greater-lactation cows. For HYKL samples, mBHB by parity group within periods 6 to 21 DIM and 22 to 60 DIM were 0.175 ± 0.001 and 0.164 ± 0.001 mmol/L for first-lactation cows; 0.169 ± 0.001 and 0.164 ± 0.001 mmol/L for second-lactation cows; and 0.180 ± 0.001 and 0.177 ± 0.001 mmol/L for third-or-greater-

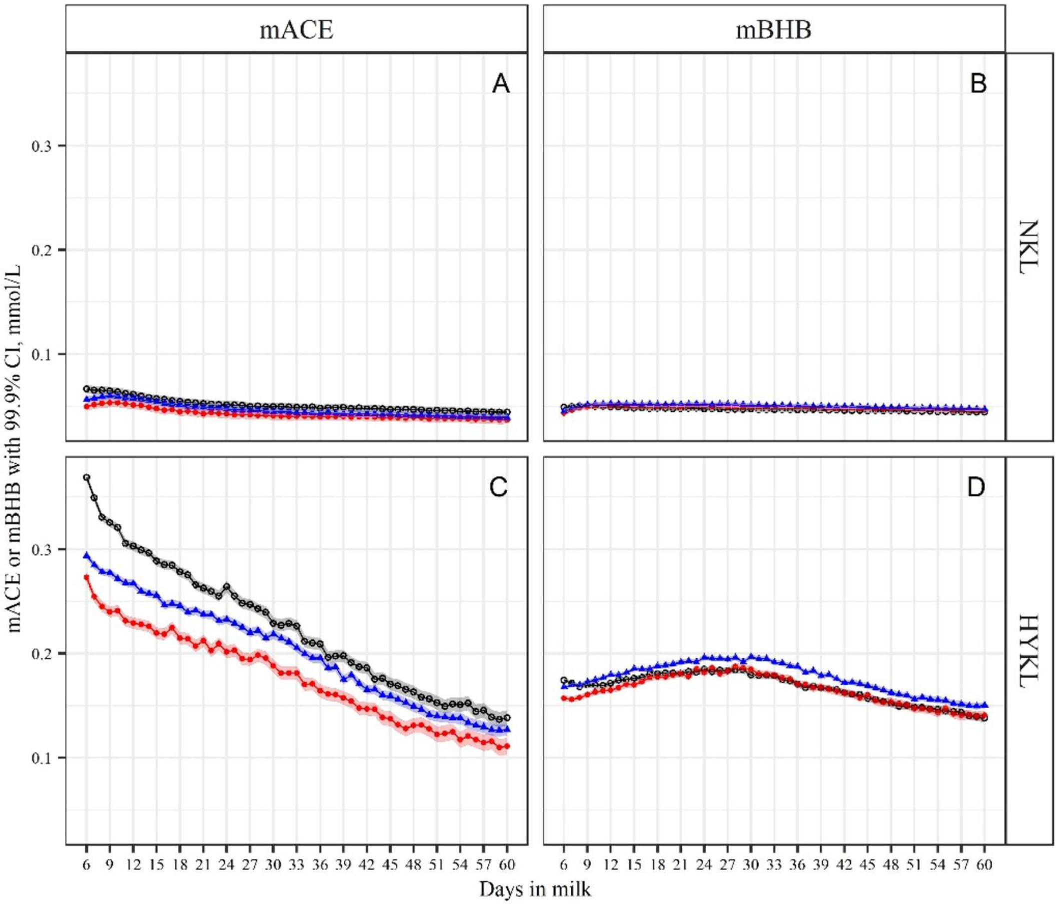

Figure 2. Least squares means of milk acetone (mACE; A and C) and β-hydroxybutyrate (mBHB; B and D) concentrations by ketolactia group (NKL; A and B, and HYKL; C and D), parity (○ 1, ● 2, ▲ ≥3), and DIM (6–60 DIM). Normoketolactia (NKL): milk samples (n = 2,616,298) with mACE <0.15 mmol/L and mBHB <0.10 mmol/L; hyperketolactia (HYKL): milk samples (n = 1,251,092) with mACE ≥0.15 mmol/L or mBHB ≥0.10 mmol/L. Final model included effects of ketolactia group ($P < 0.001$), DIM ($P < 0.001$), and all their interactions (all $P < 0.001$). Shaded area represents 99.9% CI for each DIM within each parity.

lactation cows. Within each parity group, mACE and mBHB concentrations were greater in HYKL compared with NKL samples (all $P < 0.001$).

Ketolactia Subpopulation Comparisons

Among all HYKL samples, 50.8, 41.3, and 7.9% were categorized as $\text{HYKL}_{\text{ACEBHB}}$, HYKL_{BHB}, and HYKL_{ACE}, respectively (Table 3). Parity, DIM, ketolactia subpopulations (NKL vs. HYKL_{ACE}, $\text{HYKL}_{\text{ACEBHB}}$, HYKL_{BHB}), and their interactions influenced mACE and mBHB concentrations (all $P < 0.001$).

Parity Effects

Within 6 to 60 DIM, 55.5, 44.5 and 50.7% of HYKL samples of first-, second- and third-or-greater-lactation cows, respectively, belonged to $\text{HYKL}_{\text{ACEBHB}}$ sample subpopulation (Figure 3A). For HYKL_{BHB} and HYKL_{ACE}, these percentages were 32.8, 48.9 and 43.2%, and 11.7, 6.6, and 6.1%, respectively (Figure 3B and 3C).

Mean mACE and mBHB in $\text{HYKL}_{\text{ACEBHB}}$ milk samples originating from first-, second-, and third-or-greater-lactation cows were 0.329 ± 0.002 and 0.214 ± 0.001, 0.292 ± 0.002 and 0.223 ± 0.002, and 0.311 ± 0.002 and 0.230 ± 0.001 mmol/L, respectively. Mean mACE and mBHB in HYKL_{BHB} milk samples originating from first-, second-, and third-or-greater-lactation cows were 0.090 ± 0.002 and 0.131 ± 0.001, 0.083 ± 0.002 and 0.132 ± 0.001, and 0.087 ± 0.002 and 0.135 ± 0.001 mmol/L, respectively. Mean comparisons for mACE and mBHB between $\text{HYKL}_{\text{ACEBHB}}$ and HYKL_{BHB} were different (all $P < 0.01$). Mean mACE and mBHB in HYKL_{ACE} milk samples originating from first-, second-, and third-or-greater-lactation cows were 0.173 ± 0.005 and 0.078 ± 0.003, 0.169 ± 0.007 and 0.079 ± 0.004,

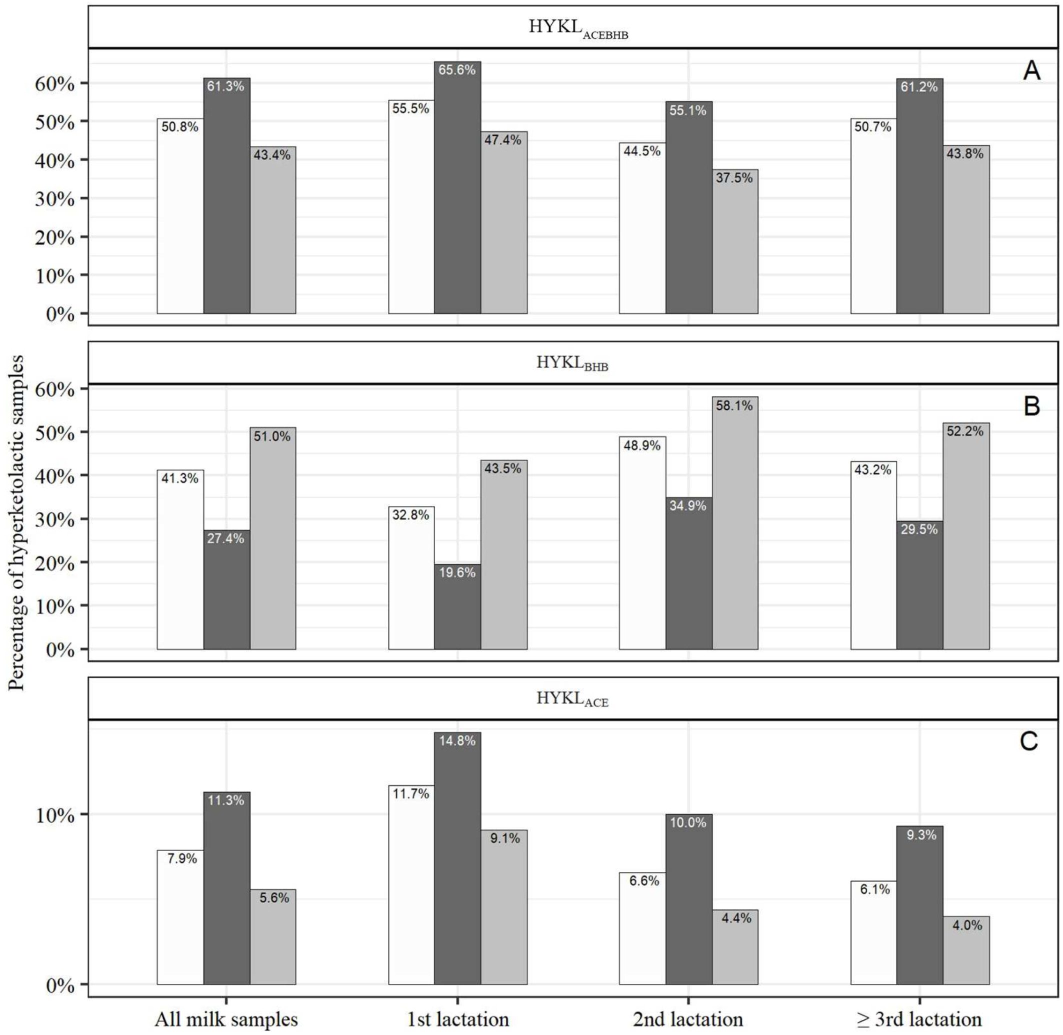

Figure 3. Percentage of hyperketolactic milk samples (HYKL: mACE ≥0.15 mmol/L or mBHB ≥0.10 mmol/L; n = 1,251,092 milk samples) by parity and stage of lactation (6–60, 6–21, and 22–60 DIM represented by white, black, and gray bars, respectively), which have elevated mBHB, mACE, or both. (A) Percentage of HYKL$_{ACEBHB}$ (mACE ≥0.15 mmol/L and mBHB ≥0.10 mmol/L; n = 635,733 milk samples); (B) percentage of HYKL$_{BHB}$ (mACE <0.15 mmol/L and mBHB ≥0.10 mmol/L; n = 516,076 milk samples); (C) percentage of HYKL$_{ACE}$ (mACE ≥0.15 mmol/L and mBHB <0.10 mmol/L; n = 99,283 milk samples). HYKL = hyperketolactia; mACE = milk acetone; mBHB = milk β-hydroxybutyrate.

and 0.172 ± 0.005 and 0.080 ± 0.003 mmol/L, respectively, and were lower compared with HYKL$_{ACEBHB}$ (all $P < 0.01$).

Time Effects

Among HYKL milk samples, there was a greater percentage of HYKL$_{ACEBHB}$ samples collected within 6 to 21 DIM than 22 to 60 DIM (61.3 vs. 43.4%, respectively, for all milk samples; Figure 3A). On the other hand, the percentage of HYKL$_{BHB}$ among HYKL samples was greater within 22 to 60 DIM than 6 to 21 DIM (51.0 vs. 27.4%, respectively, for all milk samples; Figure 3B). Percentage of HYKL$_{ACE}$ samples among all HYKL samples was greater during 6 to 21 DIM than 22 to 60 DIM (Figure 3C).

Within HYKL subpopulations, distribution patterns of mACE and mBHB over time among HYKL$_{ACEBHB}$ samples (Figure 4) were similar to those found for all HYKL samples (Figure 2). Among HYKL$_{ACE}$ or HYKL$_{BHB}$ samples, their concentrations were relatively stable over threshold values, which define these subpopulations, though mACE and mBHB concentrations declined with increasing DIM. Dynamics of mBHB concentration over DIM in HYKL$_{BHB}$ samples was best fit with a quadratic model (Table 2). Within HYKL$_{ACE}$

subpopulation samples, samples from lactations 2 and ≥3 showed the lowest mBHB quadratic fit (R^2 = 0.60) compared with regression fits for mBHB, being between 0.81 and 0.91 for other subpopulation parity samples. In contrast, mACE changes over DIM were approximately equal in R^2 fits with linear or quadratic regression models for $HYKL_{ACEBHB}$ and $HYKL_{BHB}$ subpopulations. This was not observed for $HYKL_{ACE}$ subpopulation samples, as the best fit for mACE was with a quadratic compared with linear model (R^2 = 0.94 vs. 0.76; Table 2).

Parity × Time Effects

Greater percentage of $HYKL_{ACEBHB}$ and $HYKL_{ACE}$ collected within 6 to 21 DIM than 22 to 60 DIM was especially evident among samples originated from primiparous cows (Figure 3A and 3C, respectively). Approximately 14.8 and 9.1% HYKL samples originated from primiparous cows and collected within 6 to 21 and 22 to 60 DIM, respectively, belonged to $HYKL_{ACE}$ subpopulation. In the second month of lactation, the percentage of $HYKL_{ACE}$ samples remained relatively stable over time among older cow (lactation ≥3) samples, but increased among primiparous ones (Figure 5).

Although the effects of parity × time interactions on mACE and mBHB within each subpopulation were significant, the distribution patterns of mACE and mBHB over time and within parities were similar to those described for the time effects. Within the time periods 6 to 21 DIM and 22 to 60 DIM, mACE and mBHB concentrations were different between each parity × subpopulation group for both periods (all $P < 0.001$). In comparing parity × subpopulation samples between periods 6 to 21 DIM and 22 to 60 DIM, mACE concentrations were different for $HYKL_{ACE}$ and $HYKL_{ACEBHB}$ groups for first- ($P < 0.001$ and 0.001) and second- ($P < 0.05$ and 0.01) lactation groups, respectively. Within the lactation ≥3 group, mACE was different between these 2 periods for $HYKL_{BHB}$ ($P < 0.05$) and $HYKL_{ACEBHB}$ ($P < 0.001$) groups. Between periods 6 to 21 DIM and 22 to 60 DIM, only $HYKL_{ACEBHB}$ mBHB concentrations were different by period within first- ($P < 0.001$), second- ($P < 0.01$), and third-or-greater- ($P < 0.001$) parity groups.

Figure 6 presents mean and quantiles (10 and 90%) of mACE and mBHB concentration over DIM in HYKL samples, which had mACE elevated ($HYKL_{ACEBHB}$ or $HYKL_{ACE}$). Samples located under the dashed line (Figure 6B) showing mBHB threshold (0.10 mmol/L) concentration represent $HYKL_{ACE}$ samples where "acetone exclusive type" hyperketolactia is present.

DISCUSSION

Findings of our study characterize mACE and mBHB concentrations as influenced by parity and time over early lactation from a large database of Polish Holstein cows. To our knowledge, it has been the first study describing mACE and mBHB concentrations based on such a large data set of approximately 4 million milk samples. Using threshold criteria for ACE and BHB concentrations in milk defined by de Roos et al. (2007), our data were categorized into ketolactic groups and subpopulations based on elevated concentrations of mACE, mBHB, or both. Further studies are required to establish a relationship between ketolactic status and cow health and performance. However, these findings may provide a starting basis for use of herd-based diagnostics.

This study was designed around 2 assumptions. First, FTIR calibrations provide reliable measurements of mACE and mBHB (Heuer et al., 2001; de Roos et al., 2007; Grelet et al., 2016). Second, milk ketone bodies concentrations are relatively well correlated with their blood levels (Enjalbert et al., 2001; Denis-Robichaud et al., 2014), considering blood ketone body concentration (i.e., BHB) as a benchmark in diagnosis of ketosis. We are conscious that correlation between blood and milk BHB concentration may vary from relatively low to high, dependent upon method of mBHB determination (Andersson, 1984; 1988; Enjalbert et al., 2001; Denis-Robichaud et al., 2014).

Measured mBHB and mACE concentrations by FTIR may be used to characterize the state of ketolactia (Emery et al., 1968) and have been introduced as practical methods for population monitoring of hyperketonemia (Santschi et al., 2016; Tatone et al., 2017; Chandler et al., 2018; Pralle and White, 2020), with the assumption that hyperketolactic cows are also hyperketonemic. Analysis of milk samples from routine milk recording has been suggested to be of limited value in diagnosing hyperketolactia or hyperketonemia in individual cows (Pralle et al., 2021), but it might be a good tool for herd monitoring, especially in diagnostic models mACE and mBHB, when concentrations are combined with cow data such as test-day variables (van Knegsel et al., 2010; van der Drift et al., 2012). In most studies, mBHB is considered as the main ketone body in milk, and mACE seems to be overlooked or even neglected (Santschi et al., 2016; Tatone et al., 2017; Pralle et al., 2021). Findings of our study suggest an intriguing role for mACE concentration for hyperketolactia and possibly hyperketonemia that requires further exploration. As associated metabolism is further explored, our data

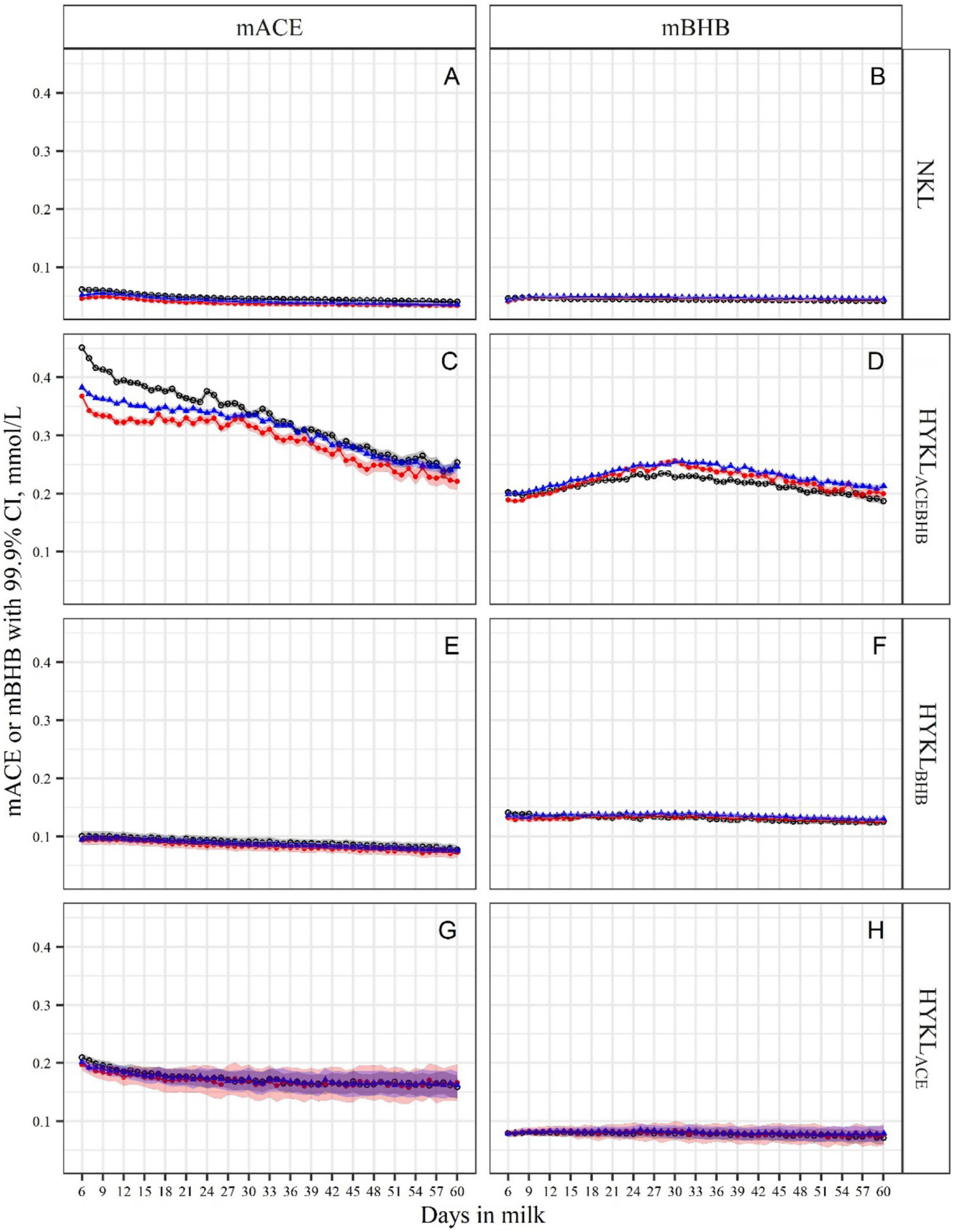

Figure 4. Least squares means of milk acetone (mACE; A, C, E, G) and β-hydroxybutyrate (mBHB; B, D, F, H) concentrations by ketolactia subpopulation (NKL: A and B; HYKL$_{ACEBHB}$: C and D; HYKL$_{BHB}$: E and F; HYKL$_{ACE}$:G and H), parity ○ 1, ● 2, ▲ ≥3) and DIM (6–60 DIM). Normoketolactic (NKL): milk samples (n = 2,616,298) with mACE <0.15 mmol/L and mBHB <0.10 mmol/L; hyperketolactic mACE and mBHB (HYKL$_{ACEBHB}$): milk samples (n = 635,733) with mACE ≥0.15 mmol/L and mBHB ≥0.10 mmol/L; hyperketolactic BHB (HYKL$_{BHB}$): milk samples (n = 516,076) with mACE <0.15 mmol/L and mBHB ≥0.10 mmol/L; hyperketolactic acetone (HYKL$_{ACE}$): milk samples (n = 99,283) with mACE ≥0.15 mmol/L and mBHB <0.10 mmol/L. Final model included effects of ketolactia subpopulation ($P < 0.001$), DIM ($P < 0.001$), and all their interactions (all $P < 0.001$). Shaded area represents 99.9% CI for each DIM within each parity.

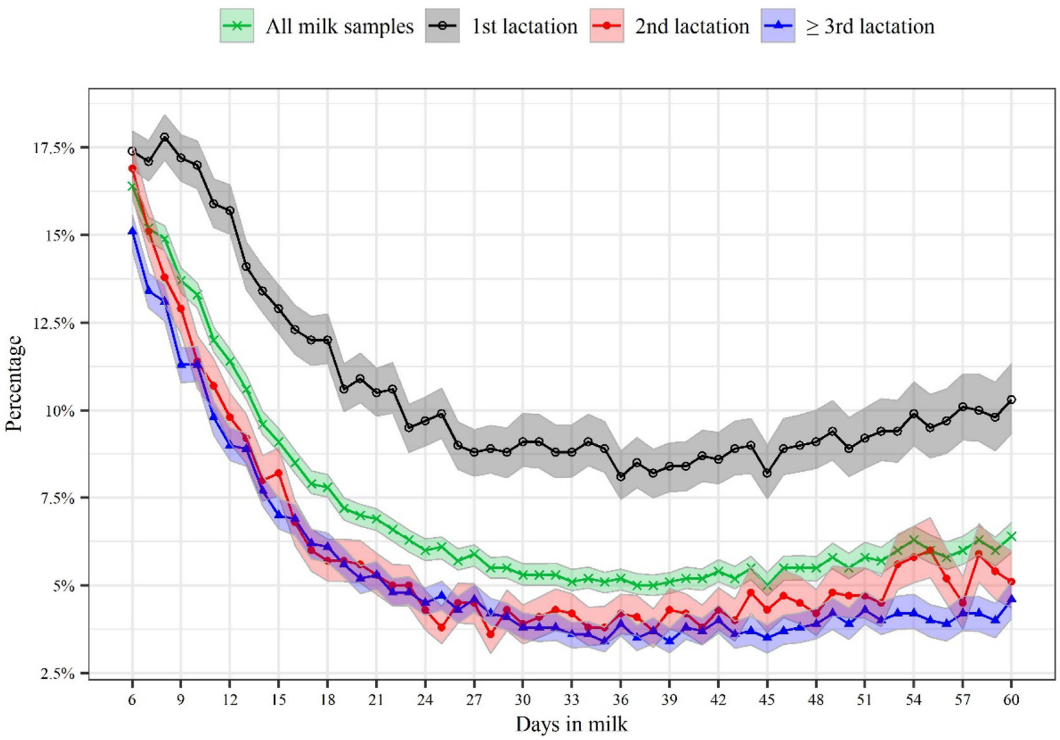

Figure 5. Percentage with 95% CI of hyperketolactia from acetone ($HYKL_{ACE}$ samples with elevated milk ACE (mACE $\geq$0.15 mmol/L) but at the same time not elevated milk BHB (mBHB <0.10 mmol/L). Milk samples (n = 99,283) originated from all, first-, second-, and $\geq$third lactation cows, and collected within the 6–60 DIM period. HYKL = hyperketolactia; mACE = milk acetone; mBHB = milk β-hydroxybutyrate.

would suggest mACE shouldn't be avoided in diagnostic or monitoring models.

Hyperketolactic Milk Samples

Milk samples considered to be HYKL were those having elevated mACE or mBHB using threshold values for mACE $\geq$0.15 mmol/L and mBHB $\geq$0.10 mmol/L, as proposed by de Roos et al. (2007). These thresholds have also been adopted by the supplier of FTIR instruments used in the current study. The number of HYKL samples found in our study (32%), assuming a direct relationship between hyperketolactia and hyperketonemia, is consistent with hyperketonemia prevalence reported by others (39% by Berge and Vertenten, 2014; 30.5% by Mann et al., 2016) who measured BHB concentration in blood. The literature review from Benedet et al. (2019) indicates hyperketonemia prevalence ranges from 11.2 to 47.2%, although method of detection, thresholds used, and biological fluid analyzed should be taken into account when interpreting these data. Santschi et al. (2016) using approximately 0.5 million samples reported 22.6% HYKL samples, which the authors described as prevalence of elevated milk BHB. Prevalence of HYKL in 3,042 Ontario herds (165,749 cows studied) was 21% (Tatone et al., 2017). In these studies, the mBHB threshold of $\geq$0.15 mmol/L was used, which may explain the observed lower prevalence compared with the current study. If higher threshold values were used, for example 0.15 mmol/L proposed by Tatone et al. (2017), 0.14 mmol/L proposed by Renaud et al. (2019), or 0.15 mmol/L proposed by Santschi et al. (2016), HYKL sample proportions in our study would be lower. However, if $HYKL_{ACE}$ samples (8%) were removed from our HYKL population, the percent of HYKL samples would be similar to these studies because they did not consider a mACE threshold and values for mBHB were only considered.

Greater HYKL samples among older cows (about 35% among third-or-greater-lactation cows) is similar to results of Santschi et al. (2016), who found a higher percentage of HYKL samples from third- compared with first-lactation cows (27.6 vs. 18.8%, respectively). Chandler et al. (2018) also found higher hyperketonemia prevalence in multiparous compared with primiparous cows, which is consistent with clinical observations of ketosis risk associated with higher milk production.

As anticipated, there were more HYKL samples among those collected within 6 to 21 DIM (43.0%) than 22 to 60 DIM (24.1%). It is worth noting the share of HYKL samples in 6 to 21 DIM was similar between primi- (43.9%) and multiparous (46.4%) cows. In contrast, older cows (lactation ≥3) had higher HYKL sample numbers in 22 to 60 DIM (32.1 vs. 24%), potentially due to energy shortage relative to high milk yield, which is more likely in multi- than primiparous cows (Hansen et al., 2006). Although classification of hyperketolactia by time postpartum (within 6–21 vs. 22–60 DIM) inspired by Oetzel's (2007) designation of type 1 and 2 ketosis may be disputable, percentage of HYKL samples in 22 to 60 DIM was much higher than that reported for ketosis by van der Drift et al. (2012) for the second month of lactation (7%). Although the diagnosis of ketosis (hyperketonemia or hyperketolactia) in the second month of lactation is not commonly performed, observations in our study may be an argument to begin doing so.

Hyperketolactic Subpopulation Milk Samples

Among defined HYKL samples, the predominant subpopulations were those having both mACE and mBHB ($HYKL_{ACEBHB}$) elevated or elevated mBHB ($HYKL_{BHB}$) but not elevated mACE, the former populations being especially abundant within 22 to 60 DIM. Although the subpopulation of samples with elevated mACE but without elevated mBHB ($HYKL_{ACE}$) was lowest, the number of such samples, especially within 6 to 21 DIM (about 11.7%), should not be neglected and requires further investigation as to potential adverse effects on cow health or performance. Among HYKL milk samples originating from primiparous cows, approximately 14.8% were considered $HYKL_{ACE}$. Assuming that mBHB concentration is correlated with blood BHB concentration (Enjalbert et al., 2001), and threshold for mBHB equal to 0.1 mmol/L correlates with blood BHB 1.2 mmol/L (van Knegsel et al., 2010; van der Drift et al., 2012), milk samples classified into $HYKL_{ACEBHB}$ and $HYKL_{BHB}$ subpopulations would be considered as obtained from hyperketonemic cows. Intuitively, $HYKL_{ACE}$ cows with excessive mACE only would not be diagnosed by blood measurement of ketone bodies (i.e., BHB). It is clearly seen that the number of such "undiagnosed" cows is higher within 6 to 21 DIM than 22 to 60 DIM, as well as much higher among primi- than multiparous cows. The metabolic underpinning of elevated mACE in primiparous cows is not known, and further research is required to understand the potential role in cow health and performance.

Milk ACE and BHB Concentrations Over Early Lactation

Parity Effect. Among all milk samples, those from second-lactation cows had the lowest mean mACE, whereas milk samples of third-or-greater-lactation cows had the highest mean mBHB. Such a distribution of mean mACE and mBHB concentrations among parities was mostly determined by the distribution among HYKL milk samples. Although parity effects on mean

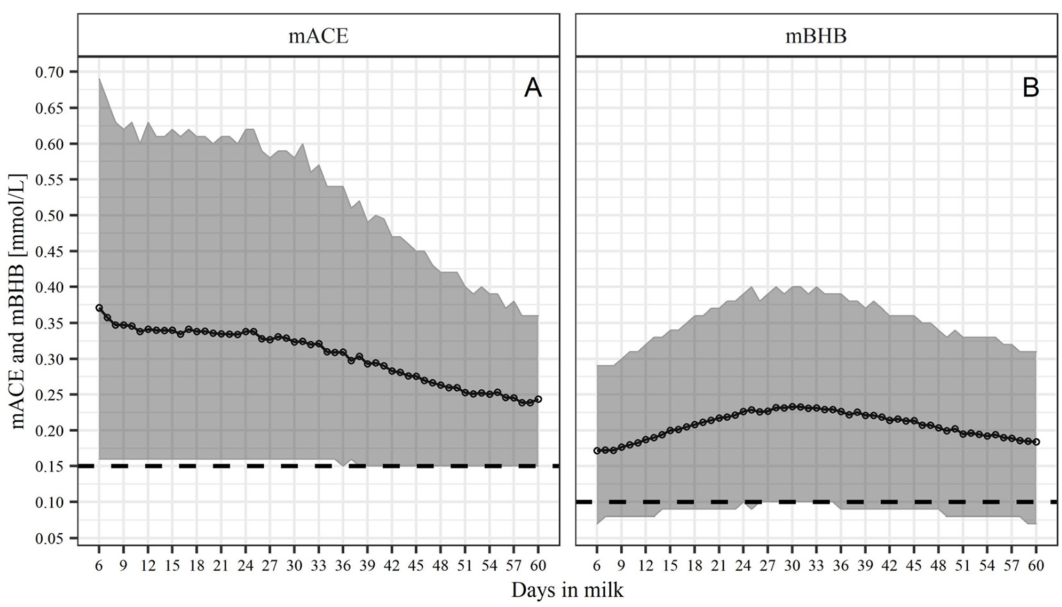

Figure 6. Mean and quantiles (10 and 90%) of milk acetone (mACE) and β-hydroxybutyrate (mBHB) concentration within the 6–60 DIM period in milk samples of hyperketolactic (HYKL) samples with mACE ≥0.15 mmol/L ($HYKL_{ACEBHB}$ or $HYKL_{ACE}$; n = 735,016 milk samples). The dashed lines show the threshold values used to define elevated mACE (0.15 mmol/|L) and mBHB (0.10 mmol/L) in HYKL cows (based on de Roos et al., 2007).

mACE or mBHB concentrations were significant among NKL samples, their biological importance is questionable. In contrast, among HYKL samples, there were clear parity differences, with highest mean mACE concentration in milk samples of primiparous cows. The differences among parities in HYKL milk samples (Figure 2) were similar to those seen in $HYKL_{ACEBHB}$ samples (Figure 4), given that these were the most abundant subpopulation of HYKL milk samples.

Time and Parity × Time Effects. Among all milk samples, irrespective of parity, mean daily mACE and mBHB concentrations declined linearly with progressing lactation. However, this declining trend was most apparent in primiparous cow milk samples collected within 6 to 21 DIM. A similar declining trend was also seen in mean mBHB concentrations for the same period. A similar pattern of distribution of mACE and mBHB over time was shown by de Roos et al. (2007), who studied the percentage of milk samples with the highest mACE (>0.290 mmol/L) and mBHB (>0.180 mmol/L) concentrations over the first 9 wk of lactation. They found highest sample percentage with elevated BHB at 3 wk, and with elevated mACE at 1 wk postpartum. Similar patterns of mACE decline during the first 9 wk of lactation were observed by van Knegsel et al. (2010), but this study was based on only 69 milk samples. A similar declining trend over time was observed for mACE among HYKL samples. Moreover, similar patterns of distribution of mean mACE and mBHB over time of early lactation were also observed within $HYKL_{ACEBHB}$ milk samples, the most abundant subpopulation among HYKL samples.

In contrast, mBHB concentration from older cow samples showed strongly quadratic responses over time, with the highest values within 10 to 30 DIM. In HYKL samples, the distribution of mean mBHB concentration over 6 to 60 DIM was similar among parities, showing a quadratic response, with peak values between 25 and 30 DIM. This observed peak in mBHB is later than reported hyperketonemia prevalence (McArt et al., 2012, 2015).

Milk in HYKL and $HYKL_{ACEBHB}$ subpopulation samples were found to have uncoordinated peaks of mean mACE and mBHB concentrations, with peak mACE occurring on 6 to 9 DIM and peak of mBHB occurring at the end of first month of lactation. Although de Roos et al. (2007) observed uncoordinated peaks of mACE and mBHB during the first weeks of lactation and commented on their existence, they did not discuss biological reasons, as well as the consequences in hyperketonemia or hyperketolactia monitoring. Only Chandler et al. (2018) mentioned that such uncoordinated peaks could be considered in practical monitoring of hyperketonemia, but they did not show in which way. As stated by van Knegsel et al. (2010), the variation in mACE and mBHB over time of early lactation may affect diagnostic value of these variables to predict risk for subclinical ketosis.

The reason for such a distribution of mBHB among $HYKL_{ACEBHB}$ samples, but not among $HYKL_{BHB}$ and $HYKL_{ACE}$ remains unknown. The mostly nonlinear relationship between mBHB and DIM in the $HYKL_{ACEBHB}$ subpopulation was of interest. In $HYKL_{ACEBHB}$ as well as $HYKL_{ACE}$ samples, peak mACE concentration in the very first weeks of lactation might be explained by greater transformation of AcAc into ACE than into BHB. The AcAc is a parent ketone body that can either be decarboxylated to ACE in a spontaneous nonenzymatic and nonreversible reaction or enzymatically reduced to BHB in reversible reaction by BHB dehydrogenase (Bergman, 1971). Moreover, a part of body BHB comes from its formation in the rumen wall (Penner et al., 2011), and day-by-day increase in DMI postpartum may enlarge total body BHB pool. A lack of appetite in the first days postpartum in cows being in a poor metabolic status (Drackley, 1999) may at least partly explain lower body BHB, and consequently lower mBHB in such cows.

Acetone Ketolactia Implications

Due to the instability of blood ACE, its determination either by laboratory methods (Työppönen and Kauppinen, 1980; Fritzsche et al., 2001) or portable hand-held meter is very difficult or impossible, respectively. There is no reliable field method of blood ACE measurement, and taking into consideration the biology of early-lactation ketone bodies formation, there would be concern in diagnosing the recognized population of hyperketonemic cows with elevated ACE but not elevated BHB. Determination of mACE, which is now available due to the FTIR technology, makes milk analysis suitable for diagnosis of such cows. This subpopulation of cows, as we calculated to be about 8% of all HYKL cows, and about 15% of primiparous HYKL cows within 6 to 21 DIM, needs to be further studied to determine if there are adverse consequences on health, production, or reproduction similar to what has been described for hyperketonemia due to elevated BHB (Duffield et al., 2009; Ospina et al., 2010; McArt et al., 2012). The negative health and production outcomes due to elevated ACE alone have been shown very rarely (Gustafsson and Emanuelson, 1996; Hanuš et al., 2017), not because they do not exist, but due to the lack of a reliable method of ACE determination. If ACE is a result of AcAc accumulation due to lipid mobilization or inflammation effects on energy metabolism, it may reflect negative effects on cow performance or health similar to BHB. Further studies are also needed to confirm that cows with elevated mACE only are in a state of ketosis, as well as if there is an association of high mACE with greater risk of elevated mBHB and potential for "classical" clinical or subclinical ketosis. Such studies may consider the following 3 potential outcomes: (1) no consequence, (2) greater risk of elevated BHB and a state of hyperketonemia as per current dogma of blood BHB being the key determinant, (3) or direct contribution to an acetone ketotic state that has similar negative attributes on cow performance as elevated BHB.

Biological explanations for differences between age of cows in number of "acetone exclusive type" of hyperketolactia is unknown. This situation raises questions on different directions of AcAc conversion into ACE and BHB in young and older cows. Is elevated ACE related to the excessive BCS prepartum (Vanholder et al., 2015), which is quite often observed in Polish replacement heifers? Are these different directions related to the lower DMI often observed pre- and postpartum in first calving cows compared with older ones (Drackley, 1999)? Although we have no biological explanation of the prevalence of "acetone exclusive type" hyperketolactia, we strongly encourage consideration of ACE body fluid content to describe ketotic status of a cow, not only for diagnosis purposes, but also in research. Our results suggest hyperketonemia diagnosis based on blood BHB measurements may underestimate the number of ketotic cows and thus incidence and prevalence of ketosis.

CONCLUSIONS

In this population study based on 3,867,390 milk samples, we have characterized mACE and mBHB concentrations during early lactation and found significant effects of parity, DIM, hyperketolactia group and subpopulation, and their interactions on mACE and mBHB concentrations. Using published criteria interpreting mACE and mBHB concentrations, a unique sample population having elevated mACE without mBHB in early lactation, especially in primiparous cows, is intriguing. Because there is no accurate cow-side field test for blood acetone measurement, milk testing for ACE may be a useful tool in such diagnostic programs in regions where ACE is determined in milk samples collected within the milk recording system. Further research is needed to determine if these samples with elevated mACE represent an unhealthy metabolic status that adversely affects cow health and performance.

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REFERENCES

Andersson, L. 1984. Concentrations of blood and milk ketone bodies, blood isopropanol and plasma glucose in dairy cows in relation to the degree of hyperketonemia and clinical signs. Zentralbl. Veterinärmed. A 31:683–693. https://doi.org/10.1111/j.1439-0442.1984.tb01327.x.

Andersson, L. 1988. Subclinical ketosis in dairy cows. Vet. Clin. North Am. Food Anim. Pract. 4:233–251. https://doi.org/10.1016/S0749-0720(15)31046-X.

Andersson, L., and U. Emanuelson. 1985. An epidemiological study of hyperketonemia in Swedish dairy cows; Determinants and the relation to fertility. Prev. Vet. Med. 3:449–462. https://doi.org/10.1016/0167-5877(85)90006-6.

Benedet, A., C. L. Manuelian, A. Zidi, M. Penasa, and M. De Marchi. 2019. Invited review: β-hydroxybutyrate concentration in blood and milk and its associations with cow performance. Animal 13:1676–1689. https://doi.org/10.1017/S175173111900034X.

Berge, A. C., and G. Vertenten. 2014. A field study to determine the prevalence, dairy herd management systems, and fresh cow clinical conditions associated with ketosis in western European dairy herds. J. Dairy Sci. 97:2145–2154. https://doi.org/10.3168/jds.2013-7163.

Bergman, E. N. 1971. Hyperketonemia-ketogenesis and ketone body metabolism. J. Dairy Sci. 54:936–948. https://doi.org/10.3168/jds.S0022-0302(71)85950-7.

Chandler, T. L., R. S. Pralle, J. R. R. Dórea, S. E. Poock, G. R. Oetzel, R. H. Fourdraine, and H. M. White. 2018. Predicting hyperketonemia by logistic and linear regression using test-day milk and performance variables in early-lactation Holstein and Jersey cows. J. Dairy Sci. 101:2476–2491. https://doi.org/10.3168/jds.2017-13209.

Clopper, C., and E. Pearson. 1934. The use of confidence or fiducial limits illustrated in the case of the binomial. Biometrika 26:404–413. https://doi.org/10.1093/biomet/26.4.404.

de Roos, A. P. W., H. J. C. M. van den Bijgaart, J. Hørlyk, and G. de Jong. 2007. Screening for subclinical ketosis in dairy cattle by Fourier transform infrared spectrometry. J. Dairy Sci. 90:1761–1766. https://doi.org/10.3168/jds.2006-203.

Denis-Robichaud, J., J. Dubuc, D. Lefebvre, and L. DesCôteaux. 2014. Accuracy of milk ketone bodies from flow-injection analysis for the diagnosis of hyperketonemia in dairy cows. J. Dairy Sci. 97:3364–3370. https://doi.org/10.3168/jds.2013-6744.

Drackley, J. K. 1999. Biology of dairy cows during the transition period: The final frontier? J. Dairy Sci. 82:2259–2273. https://doi.org/10.3168/jds.S0022-0302(99)75474-3.

Duffield, T. F., K. D. Lissemore, B. W. McBride, and K. E. Leslie. 2009. Impact of hyperketonemia in early lactation dairy cows on health and production. J. Dairy Sci. 92:571–580. https://doi.org/10.3168/jds.2008-1507.

Emery, R. S., J. W. Bell, and J. W. Thomas. 1968. Benefits obtained from the routine testing for milk ketones. J. Dairy Sci. 51:867–868. https://doi.org/10.3168/jds.S0022-0302(68)87095-X.

Enjalbert, F., M. C. Nicot, C. Bayourthe, and R. Moncoulon. 2001. Ketone bodies in milk and blood of dairy cows: Relationship between concentrations and utilization for detection of subclinical ketosis. J. Dairy Sci. 84:583–589. https://doi.org/10.3168/jds.S0022-0302(01)74511-0.

Fritzsche, I., P. Bührdel, R. Melcher, and H. J. Böhme. 2001. Stability of ketone bodies in serum in dependence on storage time and storage temperature. Clin. Lab. 47:399–403.
Gilmour, A. R., B. J. Gogel, B. R. Cullis, S. J. Welham, and R. Thompson. 2015. ASReml User Guide Release 4.1 Functional Specification. Accessed Sep. 1, 2021. www.vsni.co.uk.
Grelet, C., C. Bastin, M. Gelè, J.-B. Davière, M. Johan, A. Werner, R. Reding, J. A. Fernandez Pierna, F. G. Colinet, P. Dardenne, N. Gengler, H. Soyeurt, and F. Dehareng. 2016. Development of Fourier transform mid-infrared calibrations to predict acetone, β-hydroxybutyrate, and citrate contents in bovine milk through a European dairy network. J. Dairy Sci. 99:4816–4825. https://doi.org/10.3168/jds.2015-10477.
Gustafsson, A. H., and U. Emanuelson. 1996. Milk acetone concentration as an indicator of hyperketonaemia in dairy cows: The critical value revised. Anim. Sci. 63:183–188. https://doi.org/10.1017/S1357729800014739.
Hansen, J. V., N. C. Friggens, and S. Højsgaard. 2006. The influence of breed and parity on milk yield and milk yield acceleration curves. Livest. Sci. 104:53–62. https://doi.org/10.1016/j.livsci.2006.03.007.
Hansen, P. W. 1999. Screening of dairy cows for ketosis by use of infrared spectroscopy and multivariate calibration. J. Dairy Sci. 82:2005–2010. https://doi.org/10.3168/jds.S0022-0302(99)75437-8.
Hanuš, O., D. Falta, M. Klimešová, E. Samková, J. Řiha, G. Chládek, P. Roubal, R. Seydlová, R. Jedelská, and J. Kopecky. 2017. Analyse of repaltionships between some milk indicators of cow energy metabolism and ketosis state. Acta Univ. Agric. Silvic. Mendel. Brun. 65:1135–1147. https://doi.org/10.11118/actaun201765041135.
Heuer, C., H. J. Luinge, E. T. Lutz, Y. H. Schukken, J. H. van der Maas, H. Wilmink, and J. P. Noordhuizen. 2001. Determination of acetone in cow milk by Fourier transform infrared spectroscopy for the detection of subclinical ketosis. J. Dairy Sci. 84:575–582. https://doi.org/10.3168/jds.S0022-0302(01)74510-9.
Iwersen, M., U. Falkenberg, R. Voigtsberger, D. Forderung, and W. Heuwieser. 2009. Evaluation of an electronic cowside test to detect subclinical ketosis in dairy cows. J. Dairy Sci. 92:2618–2624. https://doi.org/10.3168/jds.2008-1795.
Kowalski, Z. M., A. Płyta, E. Rybicka, W. Jagusiak, and K. Słoniewski. 2015. Novel model of monitoring of subclinical ketosis in dairy herds in Poland based on monthly milk recording and estimation of ketone bodies in milk by FTIR spectroscopy. Pages 25–30 in Proceedings of the Technical Meeting. ICAR.
Mann, S., D. V. Nydam, A. L. Lock, T. R. Overton, and J. A. A. McArt. 2016. Association of milk fatty acids with early lactation hyperketonemia and elevated concentration of nonesterified fatty acids. J. Dairy Sci. 99:5851–5857. https://doi.org/10.3168/jds.2016-10920.
McArt, J. A. A., D. V. Nydam, and G. R. Oetzel. 2012. Epidemiology of subclinical ketosis in early lactation dairy cattle. J. Dairy Sci. 95:5056–5066. https://doi.org/10.3168/jds.2012-5443.
McArt, J. A. A., D. V. Nydam, and M. W. Overton. 2015. Hyperketonemia in early lactation dairy cattle: A deterministic estimate of component and total cost per case. J. Dairy Sci. 98:2043–2054. https://doi.org/10.3168/jds.2014-8740.
Oetzel, G. R. 2007. Herd-level ketosis – diagnosis and risk factors. Preconference Seminar 7C: Dairy Herd Problem Investigation Strategies: Transition Cow Troubleshooting. Pages 67–91 in Proc. Am. Assoc. Bovine Practitioners. American Association of Bovine Practitioners.
Ospina, P. A., D. V. Nydam, T. Stokol, and T. R. Overton. 2010. Associations of elevated nonesterified fatty acids and beta-hydroxybutyrate concentrations with early lactation reproductive performance and milk production in transition dairy cattle in the Northeastern United States. J. Dairy Sci. 93:1596–1603. https://doi.org/10.3168/jds.2009-2852.
Penner, G. B., M. A. Steele, J. R. Aschenbach, and B. W. McBride. 2011. Ruminant nutrition symposium: Molecular adaptation of ruminal epithelia to highly fermentable diets. J. Anim. Sci. 89:1108–1119. https://doi.org/10.2527/jas.2010-3378.
Pralle, R. S., J. D. Amdall, R. H. Fourdraine, G. R. Oetzel, and H. M. White. 2021. Hyperketonemia predictions provide an on-farm management tool with epidemiological insights. Animals (Basel) 11:1291. https://doi.org/10.3390/ani11051291.
Pralle, R. S., and H. M. White. 2020. Big data, big predictions: Utilizing milk Fourier-transform infrared and genomics to improve hyperketonemia management. J. Dairy Sci. 103:3867–3873. https://doi.org/10.3168/jds.2019-17379.
Renaud, D. L., D. F. Kelton, and T. F. Duffield. 2019. Validation of a test-day milk test for β-hydroxybutyrate for identifying cows with hyperketonemia. J. Dairy Sci. 102:1589–1593. https://doi.org/10.3168/jds.2018-14778.
Santschi, D. E., R. Lacroix, J. Durocher, M. Duplessis, R. K. Moore, and D. M. Lefebvre. 2016. Prevalence of elevated milk β-hydroxybutyrate concentrations in Holstein cows measured by Fourier-transform infrared analysis in dairy herd improvement milk samples and association with milk yield and components. J. Dairy Sci. 99:9263–9270. https://doi.org/10.3168/jds.2016-11128.
Stevens, W. L. 1950. Fiducial limits of the parameter of a discontinuous distribution. Biometrika 37:117–129. https://doi.org/10.1093/biomet/37.1-2.117.
Suthar, V. S., J. Canelas-Raposo, A. Deniz, and W. Heuwieser. 2013. Prevalence of subclinical ketosis and relationships with postpartum diseases in European dairy cows. J. Dairy Sci. 96:2925–2938. https://doi.org/10.3168/jds.2012-6035.
Tatone, E. H., T. F. Duffield, S. J. LeBlanc, T. J. DeVries, and J. L. Gordon. 2017. Investigating the within-herd prevalence and risk factors for ketosis in dairy cattle in Ontario as diagnosed by the test-day concentration of β-hydroxybutyrate in milk. J. Dairy Sci. 100:1308–1318. https://doi.org/10.3168/jds.2016-11453.
Työppönen , J., and K. Kauppinen. 1980. The stability and automatic determination of ketone bodies in blood samples taken in field conditions. Acta Vet. Scand. 21:55–61. https://doi.org/10.1186/BF03546900.
van der Drift, S. G. A., R. Jorritsma, J. T. Schonewille, H. M. Knijn, and J. A. Stegeman. 2012. Routine detection of hyperketonemia in dairy cows using Fourier transform infrared spectroscopy analysis of β-hydroxybutyrate and acetone in milk in combination with test-day information. J. Dairy Sci. 95:4886–4898. https://doi.org/10.3168/jds.2011-4417.
van Knegsel, A. T. M., S. G. A. van der Drift, M. Horneman, A. P. W. de Roos, B. Kemp, and E. A. M. Graat. 2010. Short communication: Ketone body concentration in milk determined by Fourier transform infrared spectroscopy: Value for the detection of hyperketonemia in dairy cows. J. Dairy Sci. 93:3065–3069. https://doi.org/10.3168/jds.2009-2847.
Vanholder, T., J. Papen, R. Bemers, G. Vertenten, and A. C. B. Berge. 2015. Risk factors for subclinical and clinical ketosis and association with production parameters in dairy cows in the Netherlands. J. Dairy Sci. 98:880–888. https://doi.org/10.3168/jds.2014-8362.
Wickham, H. 2016. ggplot2: Elegant graphics for data analysis. Springer-Verlag.

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